

*Proof.* Let  $G$  be any  $2k$ -regular graph ( $k \geq 1$ ), without loss of generality connected. By Theorem 1.8.1,  $G$  contains an Euler tour  $v_0 e_0 \dots e_{\ell-1} v_\ell$ , with  $v_\ell = v_0$ . We replace every vertex  $v$  by a pair  $(v^-, v^+)$ , and every edge  $e_i = v_i v_{i+1}$  by the edge  $v_i^+ v_{i+1}^-$  (Fig. 2.1.4). The resulting bipartite graph  $G'$  is  $k$ -regular, so by Corollary 2.1.3 it has a 1-factor. Collapsing every vertex pair  $(v^-, v^+)$  back into a single vertex  $v$ , we turn this 1-factor of  $G'$  into a 2-factor of  $G$ .  $\square$  (1.8.1)

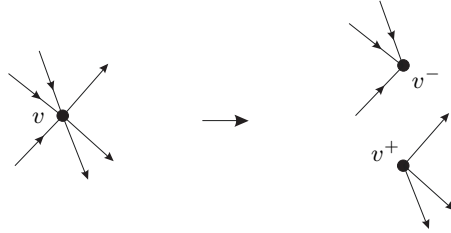


Fig. 2.1.4. Splitting vertices in the proof of Corollary 2.1.5

## 2.2 Matching in general graphs

Given a graph  $G$ , let us denote by  $\mathcal{C}_G$  the set of its components, and by  $q(G)$  the number of its *odd components*, those of odd order. If  $G$  has a 1-factor, then clearly

$$q(G - S) \leq |S| \quad \text{for all } S \subseteq V(G),$$

$\mathcal{C}_G$   
 $q(G)$   
*Tutte's condition*

since every odd component of  $G - S$  will send a factor edge to  $S$ .

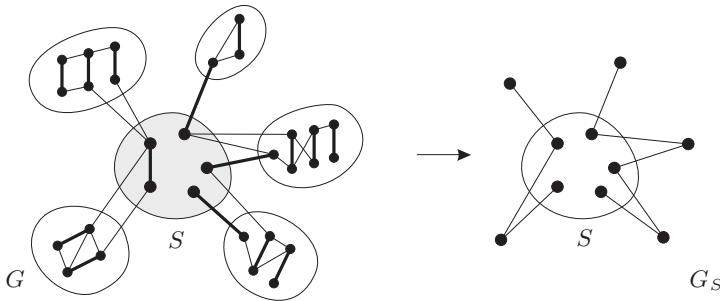


Fig. 2.2.1. Tutte's condition  $q(G - S) \leq |S|$  for  $q = 3$ , and the contracted graph  $G_S$  from Theorem 2.2.3.

Again, this obvious necessary condition for the existence of a 1-factor is also sufficient:

**Theorem 2.2.1.** (Tutte 1947)

A graph  $G$  has a 1-factor if and only if  $q(G - S) \leq |S|$  for all  $S \subseteq V(G)$ .

$V, E$   
bad set

*Proof.* Let  $G = (V, E)$  be a graph without a 1-factor. Our task is to find a bad set  $S \subseteq V$ , one that violates Tutte's condition.

We may assume that  $G$  is edge-maximal without a 1-factor. Indeed, if  $G'$  is obtained from  $G$  by adding edges and  $S \subseteq V$  is bad for  $G'$ , then  $S$  is also bad for  $G$ : any odd component of  $G' - S$  is the union of components of  $G - S$ , and one of these must again be odd.

What does  $G$  look like? Clearly, if  $G$  contains a bad set  $S$  then, by its edge-maximality and the trivial forward implication of the theorem,

*all the components of  $G - S$  are complete and every vertex  $s \in S$  is adjacent to all the vertices of  $G - s$ .* (\*)

But also conversely, if a set  $S \subseteq V$  satisfies (\*) then either  $S$  or the empty set must be bad: if  $S$  is not bad we can join the odd components of  $G - S$  disjointly to  $S$  and pair up all the remaining vertices – unless  $|G|$  is odd, in which case  $\emptyset$  is bad.

$S$   
 $a, b, c$   
 $d$   
 $M_1, M_2$

So it suffices to prove that  $G$  has a set  $S$  of vertices satisfying (\*). Let  $S$  be the set of vertices that are adjacent to every other vertex. If this set  $S$  does not satisfy (\*), then some component of  $G - S$  has non-adjacent vertices  $a, a'$ . Let  $a, b, c$  be the first three vertices on a shortest  $a$ - $a'$  path in this component; then  $ab, bc \in E$  but  $ac \notin E$ . Since  $b \notin S$ , there is a vertex  $d \in V$  such that  $bd \notin E$ . By the maximality of  $G$ , there is a matching  $M_1$  of  $V$  in  $G + ac$ , and a matching  $M_2$  of  $V$  in  $G + bd$ .

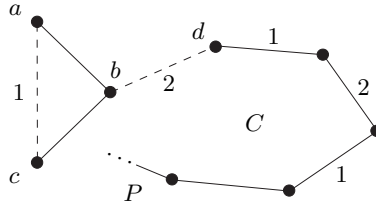


Fig. 2.2.2. Deriving a contradiction if  $S$  does not satisfy (\*)

$v$

Let  $P = d \dots v$  be a maximal path in  $G$  starting at  $d$  with an edge from  $M_1$  and containing alternately edges from  $M_1$  and  $M_2$  (Fig. 2.2.2). If the last edge of  $P$  lies in  $M_1$ , then  $v = b$ , since otherwise we could continue  $P$ . Let us then set  $C := P + bd$ . If the last edge of  $P$  lies in  $M_2$ , then by the maximality of  $P$  the  $M_1$ -edge at  $v$  must be  $ac$ , so  $v \in \{a, c\}$ ; then let  $C := dPvbd$ . This is again a cycle, since  $b$  cannot be an inner vertex of  $P$ : those are incident with  $M_2$ -edges from  $G$ , which  $b$  is not.

In every case  $C$  is an even cycle with every other edge in  $M_2$ , and whose only edge not in  $E$  is  $bd$ . Replacing in  $M_2$  its edges on  $C$  with the edges of  $C - M_2$ , we obtain a matching of  $V$  in  $E$ , a contradiction.  $\square$

**Corollary 2.2.2.** (Petersen 1891)

*Every bridgeless cubic graph has a 1-factor.*

*Proof.* We show that any bridgeless cubic graph  $G$  satisfies Tutte's condition. Let  $S \subseteq V(G)$  be given, and consider an odd component  $C$  of  $G - S$ . Since  $G$  is cubic, the degrees (in  $G$ ) of the vertices in  $C$  sum to an odd number, but only an even part of this sum arises from edges of  $C$ . So  $G$  has an odd number of  $S$ - $C$  edges, and therefore has at least 3 such edges (since  $G$  has no bridge). The total number of edges between  $S$  and  $G - S$  thus is at least  $3q(G - S)$ . But it is also at most  $3|S|$ , because  $G$  is cubic. Hence  $q(G - S) \leq |S|$ , as required.  $\square$

In order to shed a little more light on the techniques used in matching theory, we now give a second proof of Tutte's theorem. In fact, we shall prove a slightly stronger result, a result that places a structure interesting from the matching point of view on an *arbitrary* graph. If the graph happens to satisfy the condition of Tutte's theorem, this structure will at once yield a 1-factor.

A non-empty graph  $G = (V, E)$  is called *factor-critical* if  $G$  has no 1-factor but for every vertex  $v \in G$  the graph  $G - v$  has a 1-factor. We call a vertex set  $S \subseteq V$  *matchable to  $\mathcal{C}_{G-S}$*  if the (bipartite<sup>2</sup>) graph  $G_S$ , which arises from  $G$  by contracting the components  $C \in \mathcal{C}_{G-S}$  to single vertices and deleting all the edges inside  $S$ , contains a matching of  $S$ . (Formally,  $G_S$  is the graph with vertex set  $S \cup \mathcal{C}_{G-S}$  and edge set  $\{sC \mid \exists c \in C : sc \in E\}$ ; see Fig. 2.2.1.)

*factor-critical*

*matchable*

$G_S$

**Theorem 2.2.3.** (Gallai 1964; Edmonds 1965)

*Every graph  $G$  has a set  $S$  of vertices with the following two properties:*

- (i)  $S$  is matchable to  $\mathcal{C}_{G-S}$ ;
- (ii) Every component of  $G - S$  is factor-critical.

*Given any such set  $S$ , the graph  $G$  contains a 1-factor if and only if  $|S| = |\mathcal{C}_{G-S}|$ .*

For any given  $G$ , the assertion of Tutte's theorem follows easily from this result. Indeed, by (i) and (ii) we have  $|S| \leq |\mathcal{C}_{G-S}| = q(G - S)$  (since factor-critical graphs have odd order); thus Tutte's condition of  $q(G - S) \leq |S|$  implies  $|S| = |\mathcal{C}_{G-S}|$ , and the existence of a 1-factor follows from the last statement of Theorem 2.2.3.

*Proof of Theorem 2.2.3.* Note first that the last assertion of the theorem follows at once from the assertions (i) and (ii): if  $G$  has a 1-factor, we have  $q(G - S) \leq |S|$  and hence  $|S| = |\mathcal{C}_{G-S}|$  as above; conversely if

(2.1.2)

<sup>2</sup> except for the – permitted – case that  $S$  or  $\mathcal{C}_{G-S}$  is empty

$|S| = |\mathcal{C}_{G-S}|$ , then the existence of a 1-factor follows straight from (i) and (ii).

We now prove the existence of a set  $S$  satisfying (i) and (ii), by induction on  $|G|$ . For  $|G| = 0$  we may take  $S = \emptyset$ . Now let  $G$  be given with  $|G| > 0$ , and assume the assertion holds for graphs with fewer vertices.

Consider the sets  $T \subseteq V(G)$  for which Tutte's condition fails worst, i.e. for which

$$d(T) := d_G(T) := q(G - T) - |T|$$

is maximum, and let  $S$  be a largest such set  $T$ . Note that  $d(S) \geq d(\emptyset) \geq 0$ .

We first show that every component  $C \in \mathcal{C}_{G-S} =: \mathcal{C}$  is odd. If  $|C|$  is even, pick a vertex  $c \in C$ , and consider  $T := S \cup \{c\}$ . As  $C - c$  has odd order it has at least one odd component, which is also a component of  $G - T$ . Therefore

$$q(G - T) \geq q(G - S) + 1 \quad \text{while} \quad |T| = |S| + 1,$$

so  $d(T) \geq d(S)$  contradicting the choice of  $S$ .

Next we prove the assertion (ii), that every  $C \in \mathcal{C}$  is factor-critical. Suppose there exist  $C \in \mathcal{C}$  and  $c \in C$  such that  $C' := C - c$  has no 1-factor. By the induction hypothesis (and the fact that, as shown earlier, for fixed  $G$  our theorem implies Tutte's theorem) there exists a set  $S' \subseteq V(C')$  with

$$q(C' - S') > |S'|.$$

Since  $|C|$  is odd and hence  $|C'|$  is even, the numbers  $q(C' - S')$  and  $|S'|$  are either both even or both odd, so they cannot differ by exactly 1. We may therefore sharpen the above inequality to

$$q(C' - S') \geq |S'| + 2,$$

giving  $d_{C'}(S') \geq 2$ . Then for  $T := S \cup \{c\} \cup S'$  we have

$$d(T) \geq d(S) - 1 - 1 + d_{C'}(S') \geq d(S),$$

where the first '-1' comes from the loss of  $C$  as an odd component and the second comes from including  $c$  in the set  $T$ . As before, this contradicts the choice of  $S$ .

It remains to show that  $S$  is matchable to  $\mathcal{C}_{G-S}$ . If not, then by the marriage theorem there exists a set  $S' \subseteq S$  that sends edges to fewer than  $|S'|$  components in  $\mathcal{C}$ . Since the other components in  $\mathcal{C}$  are also components of  $G - (S \setminus S')$ , the set  $T = S \setminus S'$  satisfies  $d(T) > d(S)$ , contrary to the choice of  $S$ .  $\square$

Let us consider once more the set  $S$  from Theorem 2.2.3, together with any matching  $M$  in  $G = (V, E)$ . As before, we write  $\mathcal{C} := \mathcal{C}_{G-S}$ . Let us denote by  $k_S$  the number of edges in  $M$  with at least one end in  $S$ , and by  $k_{\mathcal{C}}$  the number of edges in  $M$  with both ends in  $G - S$ . Since each  $C \in \mathcal{C}$  is odd, at least one of its vertices is not incident with an edge of the second type. Therefore every matching  $M$  satisfies

$$k_S \leq |S| \quad \text{and} \quad k_{\mathcal{C}} \leq \frac{1}{2}(|V| - |S| - |\mathcal{C}|). \quad (1)$$

Moreover,  $G$  contains a matching  $M_0$  with equality in both cases: first choose  $|S|$  edges between  $S$  and  $\bigcup \mathcal{C}$  according to (i), and then use (ii) to find a suitable set of  $\frac{1}{2}(|\mathcal{C}| - 1)$  edges in every component  $C \in \mathcal{C}$ . This matching  $M_0$  thus has exactly

$$|M_0| = |S| + \frac{1}{2}(|V| - |S| - |\mathcal{C}|) \quad (2)$$

edges.

Now (1) and (2) together imply that every matching  $M$  of maximum cardinality satisfies both parts of (1) with equality: by  $|M| \geq |M_0|$  and (2),  $M$  has at least  $|S| + \frac{1}{2}(|V| - |S| - |\mathcal{C}|)$  edges, which implies by (1) that neither of the inequalities in (1) can be strict. But equality in (1), in turn, implies that  $M$  has the structure described above: by  $k_S = |S|$ , every vertex  $s \in S$  is the end of an edge  $st \in M$  with  $t \in G - S$ , and by  $k_{\mathcal{C}} = \frac{1}{2}(|V| - |S| - |\mathcal{C}|)$  exactly  $\frac{1}{2}(|\mathcal{C}| - 1)$  edges of  $M$  lie in  $C$ , for every  $C \in \mathcal{C}$ . Finally, since these latter edges miss only one vertex in each  $C$ , the ends  $t$  of the edges  $st$  above lie in different components  $C$  for different  $s$ .

The seemingly technical Theorem 2.2.3 thus hides a wealth of structural information: it contains the essence of a detailed description of all maximum-cardinality matchings in all graphs. A reference to the full statement of this result, the *Gallai-Edmonds structure theorem*, is given in the notes at the end of this chapter.

## 2.3 The Erdős-Pósa theorem

Much of the charm of König's and Hall's theorems in Section 2.1 lies in the fact that they guarantee the existence of the desired matching as soon as some obvious obstruction does not occur. In König's theorem, we can find  $k$  independent edges in our graph unless we can cover all its edges by fewer than  $k$  vertices (in which case it is obviously impossible).

More generally, if  $G$  is an arbitrary graph, not necessarily bipartite, and  $\mathcal{H}$  is any class of graphs, we might compare the largest number  $k$  of graphs from  $\mathcal{H}$  (not necessarily distinct) that we can pack disjointly into  $G$  with the smallest number  $s$  of vertices of  $G$  that will cover all